



## Lecture # 4

# Project Management and PMI

## ENGINEERING PROJECT MANAGEMENT

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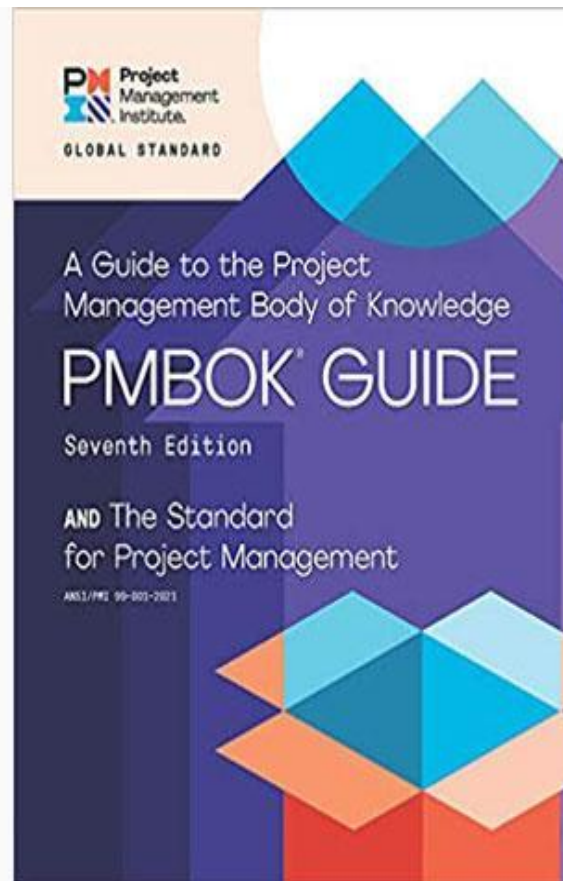
NUST COLLEGE OF E&ME

# Project Management Institute

- ▶ Project Management Institute (PMI) is the leading professional association for project management, and the authority for a growing global community of millions of project professionals and individuals who use project management skills.
- ▶ Collectively, these professionals and “change makers” consistently create better outcomes for businesses, community and society worldwide.

# PMBOK – PM Body of Knowledge

## 7<sup>th</sup> Edition



# PMI Certifications

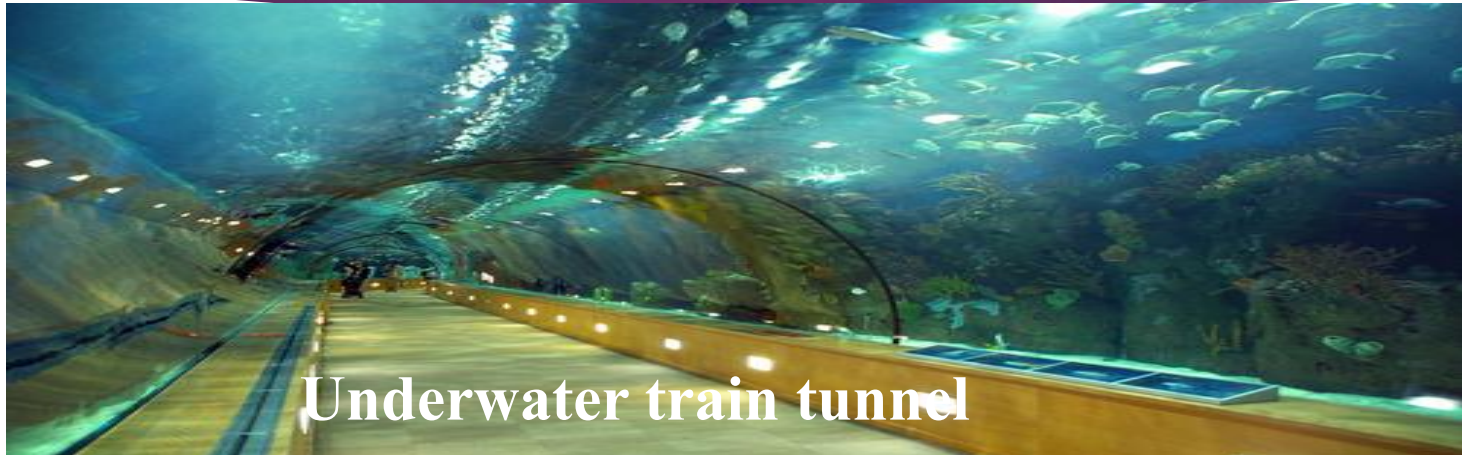
- ▶ Project Management Professional (PMP)
- ▶ Certified Associate in Project Management (CAPM)
- ▶ PMI Risk Management Professional (PMI-RMP)

# Main Features of a Project

- ▶ **Projects are Temporary**
  - ▶ Temporary means that every project has a definite beginning and a definite end.
- ▶ **Product of a Project is Unique**
  - ▶ Projects involve doing something that has not been done before and which is, therefore, *unique*.
- ▶ **Progressive Elaboration**
  - ▶ *Progressively* means "proceeding in steps; continuing steadily by increments," while *elaborated* means "worked out with care and detail; developed thoroughly"



# Examples

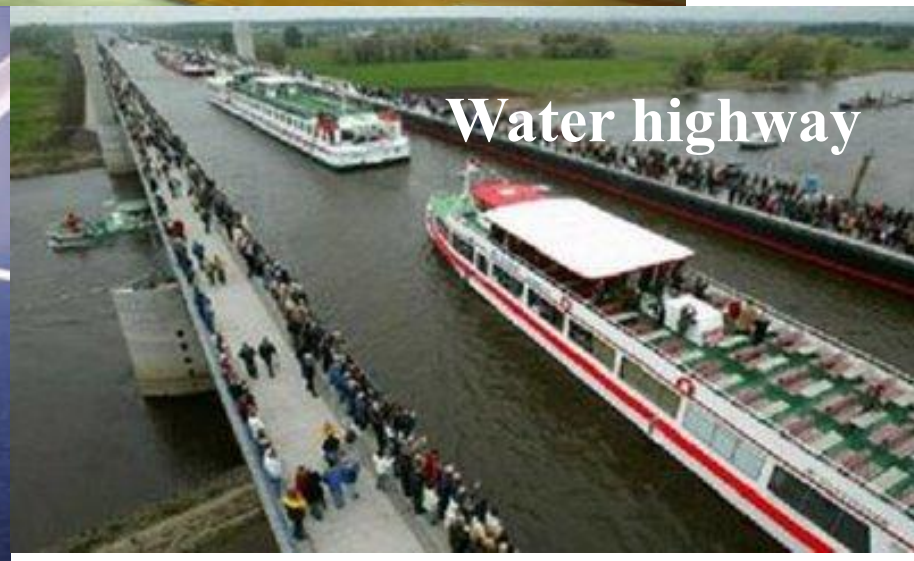


Underwater train tunnel



Wind farms in the  
middle of ocean

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Water highway

# Project Objectives

A project is a series of activities leading to achieve a specific outcome  
Within a set budget and time scale.

## Project Scope

The work that needs to be **accomplished** to deliver a product, service, or result  
with the specified features and functions.

# Project Schedule

A project schedule is a **time frame** of start and completion dates of activities constituting the project. Project scheduling looks at which tasks need to be performed for a project and assigns deadlines for their completion.

## Project Cost

A project cost is an estimate that enables the organization to evaluate the **feasibility** of the project and to manage the **capital requirements** of the project.



# Project Triple Constraints



# Operations and Projects

Two may overlap yet they share many characteristics like:

- Performed by people
- Constrained by limited resources
- Planned, executed, and controlled

Operations are ongoing and repetitive while projects are temporary and unique.

# Management ?



# Management

- ▶ Art of getting resources together to accomplish desired goals.
- ▶ Manages deployment and manipulation of human resources, financial resources, technological resources and natural resources

# Management Functions

## planning

A process that includes defining goals, establishing strategy, and developing plans to coordinate activities.

## organizing

Determining what tasks are to be done, who is to do them, how the tasks are to be grouped, who reports to whom, and where decisions are to be made.

## controlling

Monitoring activities to ensure they are being accomplished as planned and correcting any significant deviations.

## leading

A function that includes motivating employees, directing others, selecting the most effective communication channels, and resolving conflicts.

**Management  
Functions**

# Management Skills

## **technical skills**

The ability to apply specialized knowledge or expertise.

## **human skills**

The ability to work with, understand, and motivate other people, both individually and in groups

## **conceptual skills**

The mental ability to analyze and diagnose complex situations.

# Assignment # 01

- ▶ Relate your FYP for the application of Project Management. Prepare a summary of the project.
- ▶ Illustrate its characteristics
  - ▶ Goal
  - ▶ Objectives
  - ▶ Specified Time
  - ▶ Budget Resource
  - ▶ Specification
- ▶ Deadline to submit a Summary: 1 week prior to mid term
- ▶ Late submission may result in deduction of marks.



# Project Management ?



# Project Management - PMI

- ▶ Discipline of planning, organizing, securing, managing, leading, and controlling resources to achieve specific goals.
- ▶ The primary challenge is to achieve all of the project goals and objectives while honoring the preconceived constraints.
- ▶ The primary constraints are **scope, time, quality and budget**.
- ▶ The secondary —and more ambitious— challenge is to optimize the allocation of necessary inputs and integrate them to meet pre-defined objectives

# Portfolio & Portfolio Management

- ▶ A portfolio is a collection of programs and/or projects and/or other work that are grouped together to facilitate effective management of that work to meet strategic business objectives.
- ▶ The projects and programs in the portfolio may not necessarily be interdependent or directly related.